

Weekly Report 4

Project title: Scot Forge

Date: 03/02/2026

Team members:

Name: Kumar Mantha Student ID: 1002233682

Name: Vandit Goel Student ID: 1002245699

Name: Pratyaksh Mathur Student ID: 1002228261

Progress report: (please indicate the contribution of each of the team members)

As discussed in our last call, we trained multiple ML models this week using different variations and training techniques. Below is a summary of our key findings:

Usage type	Model	RMSE	MAE	MAPE	Description	
Total Usage	Xgboost	270.5	203.77	17%	Basic Xgboost	Kumar
	LGBM	262.14	194.48	19%	Basic LGBM	Kumar
	LGBM	271.12	198.79	20%	LGBM + Regime based training	Kumar
	Catboost	258.57	191.82	20%	Basic Catboost	Kumar
	Ensemble	265.16	192.59	21%	Average ensemble of Xgboost,lgbm and catboost	Kumar
	Catboost	258	191.77	19%	CatBoost Regime + Residual + Hyperparameter Tuning	Kumar
	LGBM	262.73	198.07	19%	LGBM +Temperature columns	Kumar

Usage 1	TimesNet	10.709	8.9209	3%	Base TimesNet (raw Forecast)	Vandit
Usage 2	TimesNet	292.17	289.17	17%	Base TimesNet (raw Forecast)	Vandit
	TimesNet	234.42	193.81	11%	Caliberated TimeNet (TimesNet + Online Correction)	Vandit
Usage 1	PatchTST	43.417	32.1	25%	Patch Time Series Transformer	Pratyaksh
Usage 2	PatchTST	255.14	184.86	20%	Patch Time Series Transformer	Pratyaksh

We used a 1-day walk-forward approach where we retrained the models each day on all historical data up to that day, predicted the next day only, and repeated this across the forecast period (2024) to simulate real production forecasting.